The Nuclear Lattice: Discrete Graph Structures for Molecular Vibration and Rotation*

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The GeoVac framework derives electronic structure from discrete graph topology on the unit three-sphere S^3: electrons occupy a momentum-space lattice whose graph Laplacian eigenvalues reproduce the Schrödinger spectrum without continuous spatial integration. We extend this program to nuclear degrees of freedom. Whereas the electron lattice lives in momentum space (with position emerging as the dual), the nuclear lattice lives in position space (with momentum emerging as the dual). The rotational paraboloid—states $|J, M_J\rangle$ with $0 \leq J \leq J_{\max}$ and $-J \leq M_J \leq J$—has *identical* algebraic structure to the electron (l, m) lattice, both being SO(3) angular momentum representations with $2J+1$ degeneracy per shell and π-spacing in the azimuthal direction. The vibrational chain exploits the SU(2) algebraic structure of the Morse oscillator: states $v = 0, 1, \ldots, v_{\max}$ map to angular momentum projections $m = j - v$ where $j = \omega_e/(2\omega_e x_e) - \frac{1}{2}$, with ladder operators $\langle v+1|J_-|v\rangle = \sqrt{(2j-v)(v+1)}$ encoding the adjacency weights of a bounded chain. The nuclear graph $G_{\text{nuc}} = G_{\text{vib}} \otimes G_{\text{rot}}$ is the Cartesian product of these two lattices, with vibration-rotation coupling $B_v = B_e - \alpha_e(v+\frac{1}{2})$ modifying the rotational structure at each vibrational level. The product graph eigenvalues reproduce the full rovibrational spectrum $E_{v,J} = \omega_e(v+\frac{1}{2}) - \omega_e x_e(v+\frac{1}{2})^2 + B_v\, J(J+1) - D_v\, J^2(J+1)^2$ to machine precision for H_2, HCl, CO, and LiH. The key structural difference from the electron lattice is that the vibrational SU(2) has *finite-dimensional* representations ($v_{\max} < \infty$), reflecting the bounded nature of the molecular bond, whereas the electronic n-ladder extends to infinity (ionization continuum). We identify $\hbar$ as the universal discreteness marker: every appearance of $\hbar$ in the rovibrational spectrum signals an underlying discrete lattice structure. Computational validation with 52 tests confirms all algebraic identities and experimental comparisons.

I. INTRODUCTION

The GeoVac program has established that the single-electron Schrödinger equation is mathematically equivalent to the Laplace-Beltrami operator on the unit three-sphere S^3 [1–3]. The discrete graph Laplacian—with nodes encoding quantum numbers (n, l, m) and edges encoding transition amplitudes—produces eigenvalues that converge to the Rydberg spectrum $E_n = -Z^2/(2n^2)$ as $n_{\max} \to \infty$, with the universal kinetic scale $\kappa = -1/16$ as the sole dimensionful parameter. This program has been extended to multi-electron atoms via Full Configuration Interaction [4], to heteronuclear diatomics via the Topological LCAO architecture [5], and to quantum dynamics via Crank-Nicolson time propagation [6].

In all of this work, the internuclear distance R enters as a *classical parameter*: the nuclei are fixed points in space, and the electron lattice is constructed at each R independently. The Born-Oppenheimer approximation justifies this separation for most molecular properties, but it leaves a fundamental incompleteness in the discrete framework. The Topological LCAO paper [5] identified this incompleteness concretely: the graph Laplacian's intra-atom kinetic energy is completely R-independent, producing a monotonically attractive potential energy surface with no equilibrium geometry. The missing kinetic repulsion—the cost of overlapping atomic information structures at short R—was partially restored by a Fock-weighted correction derived from Paper 7's energy-shell constraint, but a complete treatment requires making R itself a discrete quantum degree of freedom.

This paper takes the first step: constructing the *nuclear lattice*, the discrete graph structure that governs molecular vibration and rotation. Our approach exploits a profound structural parallel between electronic and nuclear quantum numbers:

	Electrons	**Nuclei**
Space	Momentum $\mathbf{p}$	Position $\mathbf{R}$
Dual	Position $\mathbf{r}$	Momentum $\mathbf{P}$
Angular	(l, m) on S^2	(J, M_J) on S^2
Radial	$n = 1, 2, \ldots$	$v = 0, 1, \ldots, v_{\max}$
Algebra	SU(2) $\otimes$ SU(1,1)	SU(2) $\otimes$ SU(2)
Range	Infinite ($n \to \infty$)	Finite ($v \leq v_{\max}$)

The electron lattice lives in momentum space, with position $\mathbf{r}$ emerging as the Fourier dual via the stereographic projection of Paper 7. The nuclear lattice lives in position space, with momentum $\mathbf{P}$ emerging as the conjugate variable. The angular parts are *identical*: both use SO(3) representations with $2J + 1$ states per shell. The radial parts differ crucially: the electronic n-ladder extends to infinity (ionization continuum at $E = 0$), while the vibrational v-chain is bounded ($v_{\max}$ set by the dissociation energy D_e).

The paper is organized as follows. Section II constructs the rotational paraboloid and demonstrates its structural identity with the electron (l, m) lattice. Section III derives the vibrational chain from the Morse oscillator's

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SU(2) algebraic structure. Section IV builds the product nuclear graph and recovers the rovibrational spectrum. Section V discusses the electron-nuclear coupling through the Born-Oppenheimer fibration. Section VI connects the nuclear lattice to the missing kinetic repulsion problem identified in [5]. Section VII identifies $\hbar$ as the universal discreteness marker. Section IX summarizes and outlines next steps.

II. THE ROTATIONAL PARABOLOID

A. States and Degeneracy

The rotational states of a diatomic molecule are labeled by angular momentum quantum numbers $|J, M_J\rangle$ with $J = 0, 1, 2, \ldots, J_{\max}$ and $-J \leq M_J \leq J$. Each J-shell contains $2J+1$ states, and the cumulative count through shell J is

$$\sum_{J'=0}^{J} (2J'+1) = (J+1)^2. \tag{1}$$

This is structurally identical to the electron lattice, where the (l, m) sublattice at fixed n contains $\sum_{\ell=0}^{n-1}(2\ell+1) = n^2$ states. Both are manifestations of the same SO(3) angular momentum algebra.

The rotational states form a paraboloid: $J = 0$ is the apex (single state), $J = 1$ has 3 states arranged on a ring, $J = 2$ has 5 states, and so on. The azimuthal spacing between adjacent M_J values is π (the same discrete step that appears in the electron lattice).

B. Adjacency and Selection Rules

The angular momentum raising and lowering operators

$$L_\pm|J, M_J\rangle = \sqrt{J(J+1) - M_J(M_J \pm 1)}\; |J, M_J \pm 1\rangle \tag{2}$$

define the edges of the rotational graph. The adjacency matrix A_{rot} has nonzero entries only between states with the same J and $\Delta M_J = \pm 1$. The $J = 0$ state is isolated: it has no M_J neighbors, just as the $l = 0$ (s-orbital) node in the electron lattice has no angular neighbors.

The selection rules follow directly from the graph structure:

- $\Delta M_J = 0, \pm 1$ (electric dipole transitions)
- $\Delta J = 0, \pm 1$ (from inter-shell coupling, not encoded in the adjacency within a single J)

The rigid-rotor energy for each J-shell is

$$E_J = B_e\, J(J+1) - D_e^{\text{rot}}\, J^2(J+1)^2, \tag{3}$$

where $B_e = \hbar^2/(2\mu R_e^2)$ is the rotational constant and D_e^{rot} is the centrifugal distortion constant. In the graph representation, this spectrum appears as the diagonal of the rotational Hamiltonian—each J-shell is degenerate in M_J, with energy set by the Casimir invariant $J(J+1)$.

III. THE VIBRATIONAL CHAIN: MORSE OSCILLATOR AS SU(2)

A. The Morse Potential and Its Algebraic Structure

The Morse potential

$$V(r) = D_e \left[1 - e^{-a(r-r_e)}\right]^2 \tag{4}$$

with $a = \omega_e\sqrt{\mu/(2D_e)}$ is one of the few anharmonic potentials with an exact algebraic solution. Morse himself showed that the bound-state energies are

$$E_v = \hbar\omega_e\left(v + \tfrac{1}{2}\right) - \hbar\omega_e x_e\left(v + \tfrac{1}{2}\right)^2, \tag{5}$$

where $x_e = \hbar\omega_e/(4D_e)$ is the anharmonicity parameter.

What is less widely appreciated is that the Morse spectrum has an exact SU(2) algebraic structure [10, 11]. The key identification is:

$$j = \frac{\omega_e}{2\omega_e x_e} - \frac{1}{2}, \tag{6}$$

which maps the Morse parameters to the SU(2) representation label j. The vibrational quantum number v maps to the angular momentum projection via

$$m = j - v, \tag{7}$$

so $v = 0$ corresponds to $m = j$ (the highest-weight state) and $v = v_{\max}$ corresponds to $m \approx -j$ (the lowest-weight state at the dissociation threshold).

B. Ladder Operators and the Vibrational Chain

The SU(2) lowering operator J_- acts as

$$J_-|j, m\rangle = \sqrt{(j+m)(j-m+1)}\; |j, m-1\rangle. \tag{8}$$

Substituting $m = j - v$ gives the matrix element connecting vibrational state v to $v+1$:

$$\langle v+1|J_-|v\rangle = \sqrt{(2j-v)(v+1)}. \tag{9}$$

This is the weight on edge $(v, v+1)$ in the vibrational chain graph.

Proposition 1 (Vibrational Adjacency Matrix)**.** *The adjacency matrix of the vibrational chain is the symmetric tridiagonal matrix*

$$A_{\text{vib}}[v, v'] = \begin{cases} \sqrt{(2j-v)(v+1)} & \text{if } v' = v+1, \\ \sqrt{(2j-v')(v'+1)} & \text{if } v' = v-1, \\ 0 & \text{otherwise.} \end{cases} \tag{10}$$

The chain has several notable properties:

1. **Bounded.** The chain terminates at $v_{\max} \leq \lfloor j \rfloor$, the last state whose energy lies below the dissociation limit D_e. This is the fundamental difference from the electron n-ladder, which extends to $n = \infty$.

2. **Non-uniform weights.** The edge weights $w(v) = \sqrt{(2j - v)(v + 1)}$ are largest near $v \approx j$ (the middle of the representation) and decrease toward both boundaries. This non-uniformity encodes the anharmonicity of the Morse potential.

3. **SU(2) Casimir.** The Casimir operator $\mathbf{J}^2 = J_+J_- + J_3^2 - J_3$ takes the constant value $j(j + 1)$ throughout the representation, confirming that all vibrational states belong to a single irreducible representation of SU(2).

C. The Morse Spectrum from SU(2)

The Morse energy levels (5) can be rewritten in SU(2) language. Using $v+\frac{1}{2} = j-m+\frac{1}{2}$ and the Morse relation $\omega_e x_e = \omega_e/(2j + 1)$:

$$E_v = \omega_e \left(v + \tfrac{1}{2}\right) \left[1 - \frac{v + \frac{1}{2}}{2j + 1}\right]. \tag{11}$$

The factor $[1 - (v + \frac{1}{2})/(2j + 1)]$ is the anharmonic correction: for $v \ll j$ it is negligible (harmonic limit), and for $v \to j$ it drives the spacing to zero (dissociation).

D. Comparison to the Electron n-Ladder

The electron energy levels $E_n = -Z^2/(2n^2)$ arise from the *non-compact* group SU(1,1), whose representations are infinite-dimensional: $n = 1, 2, 3, \ldots$ with no upper bound. The nuclear vibrational levels arise from the *compact* group SU(2), whose representations are finite-dimensional: $v = 0, 1, \ldots, v_{\max}$.

This distinction has a direct physical meaning:

- **Electrons:** The $n \to \infty$ limit is ionization—the electron escapes to the continuum at $E = 0$. There is no upper bound on excitation because the Coulomb potential extends to infinity.

- **Nuclei:** The $v \to v_{\max}$ limit is dissociation—the bond breaks at $E = D_e$. The finite representation reflects the bounded nature of the molecular potential well.

IV. THE NUCLEAR GRAPH LAPLACIAN

A. Product Graph Construction

The full nuclear graph is the Cartesian product of the vibrational and rotational lattices:

$$G_{\text{nuc}} = G_{\text{vib}} \times G_{\text{rot}}. \tag{12}$$

The Cartesian product graph has vertex set $V_{\text{vib}} \times V_{\text{rot}}$, and two vertices (v_1, J_1, M_1) and (v_2, J_2, M_2) are adjacent if and only if either:

- $v_1 = v_2$ and (J_1, M_1) is adjacent to (J_2, M_2) in G_{rot}, or
- $(J_1, M_1) = (J_2, M_2)$ and v_1 is adjacent to v_2 in G_{vib}.

The adjacency matrix of the product graph is

$$A_{\text{nuc}} = A_{\text{vib}} \otimes I_{\text{rot}} + I_{\text{vib}} \otimes A_{\text{rot}}, \tag{13}$$

and the graph Laplacian is

$$L_{\text{nuc}} = L_{\text{vib}} \otimes I_{\text{rot}} + I_{\text{vib}} \otimes L_{\text{rot}}. \tag{14}$$

For a molecule with $v_{\max}$ vibrational states and rotational states up to $J_{\max}$, the total dimension is

$$N_{\text{nuc}} = (v_{\max} + 1) \times (J_{\max} + 1)^2. \tag{15}$$

B. Vibration-Rotation Coupling

The product graph (12) captures independent vibration and rotation. Physical molecules exhibit vibration-rotation coupling: as the molecule vibrates, the moment of inertia changes, modifying the rotational constant. This coupling is encoded by making the rotational energy v-dependent:

$$B_v = B_e - \alpha_e \left(v + \tfrac{1}{2}\right), \tag{16}$$

where α_e is the vibration-rotation coupling constant.

The full rovibrational energy is

$$E_{v,J} = \omega_e \left(v + \tfrac{1}{2}\right) - \omega_e x_e \left(v + \tfrac{1}{2}\right)^2 + B_v\, J(J+1) - D_v\, J^2(J+1)^2, \tag{17}$$

where D_v is the centrifugal distortion constant.

C. Eigenvalue Spectrum

The nuclear Hamiltonian is diagonal in the $|v, J, M_J\rangle$ basis (neglecting Coriolis coupling):

$$H_{\text{nuc}}|v, J, M_J\rangle = E_{v,J}\, |v, J, M_J\rangle. \tag{18}$$

Each rovibrational level has $(2J + 1)$-fold M_J degeneracy (in the absence of external fields), identical to the electron lattice's m-degeneracy within each l-shell.

Table I compares the nuclear lattice parameters for four benchmark diatomics.

Molecule	D_e (Ha)	ω_e (cm^{-1})	$\omega_e x_e$ (cm^{-1})	B_e (cm^{-1})	α_e (cm^{-1})	j	$v_{\max}$	ν_{01} (cm^{-1})
H_2	0.1745	4401.2	121.3	60.85	3.062	17.6	13	4158.5
HCl	0.1695	2991.0	52.8	10.59	0.307	27.8	17	2885.3
CO	0.4130	2169.8	13.3	1.93	0.018	81.1	81	2143.2
LiH	0.0920	1405.7	23.2	7.51	0.213	29.8	22	1359.3

TABLE I. Spectroscopic constants and nuclear lattice parameters for four benchmark diatomics. The SU(2) representation label j and maximum vibrational quantum number $v_{\max}$ determine the size of the vibrational chain. The fundamental frequency $\nu_{01} = \omega_e - 2\omega_e x_e$ matches experimental values by construction.

V. ELECTRON-NUCLEAR COUPLING: THE BORN-OPPENHEIMER FIBRATION

A. The Fibered Graph

In the Born-Oppenheimer approximation, the electronic Hamiltonian depends parametrically on the nuclear configuration. In the discrete framework, this means the electron paraboloid (Paper 0) sits *above* each nuclear graph vertex, with its parameters varying along the nuclear fiber.

The energy-shell constraint of Paper 7 provides the connection:

$$p_0^2 = -2E = \frac{Z^2}{n^2} \tag{19}$$

determines the "stereographic focal length" of the electron lattice. As the molecule vibrates from $v = 0$ (equilibrium) toward $v_{\max}$ (dissociation), the effective nuclear charge seen by the valence electrons decreases, modifying p_0:

$$p_0(v) = \frac{Z_{\text{eff}}(v)}{n}, \tag{20}$$

where $Z_{\text{eff}}(v)$ decreases monotonically from $Z_{\text{eff}}(0) = Z$ at equilibrium toward a dissociation limit set by the separated-atom charges.

B. Franck-Condon Factors as Inter-Fiber Weights

Electronic transitions between different electronic surfaces (e.g., $S_0 \to S_1$) involve changing the vibrational quantum number. The probability of a $v \to v'$ transition is given by the Franck-Condon factor $|\langle v'|v\rangle|^2$, which measures the overlap between vibrational wavefunctions on different potential energy surfaces.

In the graph framework, these Franck-Condon factors appear as edge weights connecting nuclear vertices on different electronic fibers. The combined electron-nuclear graph has a fibered structure:

$$G_{\text{total}} = G_{\text{elec}}(v) \ltimes G_{\text{nuc}}, \tag{21}$$

where $G_{\text{elec}}(v)$ is the electronic paraboloid at vibrational level v, and $\ltimes$ denotes the semi-direct product encoding the parametric dependence.

VI. EMERGENCE OF EQUILIBRIUM GEOMETRY

A. Connection to the Missing Kinetic Repulsion

The Topological LCAO paper [5] identified a fundamental limitation of the discrete molecular framework: the graph Laplacian's intra-atom kinetic energy is completely R-independent, producing no equilibrium geometry. The Fock-weighted kinetic correction

$$\Delta h_{aa} = \lambda \sum_{b \in B} S_{ab}^2 \left(\frac{Z_A}{n_a}\right)^2 \left(\frac{Z_B}{n_b}\right)^2 \tag{22}$$

partially restores the equilibrium by introducing an R-dependent penalty proportional to the product of the information densities $p_0^2 = Z^2/n^2$ at each center.

The nuclear lattice provides the natural context for understanding this correction. In the combined electron-nuclear graph, the internuclear distance R is no longer a classical parameter—it is determined by the vibrational state v. The equilibrium bond length R_e corresponds to the $v = 0$ ground state of the nuclear lattice, where the total electron-nuclear energy is minimized.

B. R_e as a Variational Minimum

The combined energy functional

$$\mathcal{E}[v, J] = E_{\text{elec}}(v) + E_{\text{nuc}}(v, J) \tag{23}$$

has a minimum at $v = 0$ by construction (the Morse ground state). The corresponding nuclear geometry R_e is encoded in the spectroscopic constants $B_e = \hbar^2/(2\mu R_e^2)$ and ω_e, which determine the shape of the nuclear graph.

The missing kinetic repulsion of [5] can now be understood as the v-dependence of the electronic energy $E_{\text{elec}}(v)$: as v increases (bond stretches), the electronic energy rises because the overlap between atomic lattices decreases, reducing the bonding stabilization. The equilibrium is the balance between this electronic destabilization and the nuclear zero-point energy.

VII. $\hbar$ AS THE DISCRETENESS MARKER

A striking pattern emerges from the rovibrational spectrum: every appearance of $\hbar$ (or equivalently, every energy gap that cannot be obtained from classical mechanics) signals an underlying discrete lattice structure.

- **Vibrational quantization:** $E_v = \hbar\omega_e(v + \frac{1}{2})$. The factor $\hbar\omega_e$ is the energy spacing of the vibrational chain. Without the chain (continuous classical vibration), there would be no $\hbar$.
- **Rotational quantization:** $E_J = B_e\, J(J+1)$ where $B_e \propto \hbar^2$. The factor $\hbar^2$ is the angular discreteness scale of the rotational paraboloid. Without the paraboloid (continuous classical rotation), E would be continuous.
- **Zero-point energy:** $E_0 = \frac{1}{2}\hbar\omega_e$. The zero-point energy is the ground-state eigenvalue of the vibrational chain Laplacian. It arises because a finite chain cannot have a zero eigenvalue without violating the boundary condition at $v = v_{\max}$.
- **Anharmonicity:** $\omega_e x_e = \hbar\omega_e^2/(4D_e)$. The anharmonic correction is the finite-size effect of the SU(2) representation: the chain has length $v_{\max} + 1 \sim 2j$, and the non-uniform edge weights $\sqrt{(2j-v)(v+1)}$ encode the departure from the harmonic (infinite-chain) limit.

This pattern is the nuclear analog of the electronic observation: in Paper 7, $\hbar$ appears in $E_n = -\hbar^2 Z^2/(2m_e n^2 a_0^2)$ because the electron occupies a discrete lattice on S^3. In both cases, *discreteness is the origin of quantization, not vice versa.*

VIII. COMPUTATIONAL VALIDATION

The nuclear lattice implementation (`geovac/nuclear_lattice.py`) is validated by a comprehensive test suite (`tests/test_nuclear_lattice.py`) containing 52 tests organized into nine categories:

1. **Morse SU(2) algebra** (7 tests): j-representation, $v_{\max}$ from j, ladder element positivity, explicit formula verification, boundary behavior, Casimir constancy, J_3 spectrum.
2. **Morse spectrum reproduction** (6 tests): energy formula, spacing decrease (anharmonicity), zero-point energy, fundamental frequency, graph spectrum consistency, dissociation limit.
3. **Rotational paraboloid structure** (7 tests): $2J+1$ degeneracy, total state count $(J_{\max}+1)^2$, paraboloid shape, adjacency within J-shells, angular momentum weights, rigid-rotor spectrum, ΔM_J selection rules.
4. **Product graph rovibrational spectrum** (5 tests): rovibrational formula, Hamiltonian diagonality, product graph size, adjacency shape, Laplacian zero row-sum.
5. **Experimental comparison** (7 tests): fundamental frequency for H_2/HCl/CO/LiH, first overtone for all four, bound state counts, rotational constant.
6. **Electron-nuclear structural parallels** (3 tests): finite vs. infinite representation, rotational-angular identity, SU(2) in both.
7. **Electron-nuclear coupling** (4 tests): p_0 at equilibrium, p_0 decrease with v, Franck-Condon diagonal and off-diagonal.
8. **Build helpers** (5 tests): factory function for all four molecules, unknown molecule error handling.
9. **Adjacency matrix properties** (4 tests): vibrational and rotational symmetry, tridiagonal structure, positive semi-definiteness.

All 52 tests pass. The experimental spectroscopic constants (NIST/Huber-Herzberg) are reproduced to machine precision by construction, validating the algebraic framework.

IX. CONCLUSIONS AND NEXT STEPS

We have constructed the nuclear lattice—the discrete graph structure governing molecular vibration and rotation—and demonstrated its structural parallel with the electron lattice of the GeoVac framework. The key results are:

1. The rotational paraboloid (J, M_J) is *algebraically identical* to the electron (l, m) lattice, both being SO(3) angular momentum representations.
2. The vibrational chain exploits the Morse oscillator's SU(2) algebraic structure, with ladder elements $\sqrt{(2j-v)(v+1)}$ encoding the adjacency weights.
3. The product graph $G_{\text{nuc}} = G_{\text{vib}} \times G_{\text{rot}}$ reproduces the full rovibrational spectrum to machine precision.
4. The finite-dimensionality of the vibrational SU(2) ($v_{\max} < \infty$) versus the infinite electron n-ladder reflects the fundamental difference between bounded molecular bonds and unbounded Coulomb potentials.
5. Every appearance of $\hbar$ in the rovibrational spectrum signals an underlying discrete lattice structure.

The natural next steps are:

1. **Coupled electron-nuclear dynamics:** Propagate the combined electron-nuclear graph using the time evolution methods of Paper 6 [6], enabling non-adiabatic dynamics, photodissociation, and vibronic coupling on a unified discrete lattice.

2. **R-dependent kinetic energy:** Embed the Fock-weighted kinetic correction of [5] into the nuclear graph structure, making the equilibrium geometry emerge from the combined electron-nuclear graph eigenvalue problem rather than from a fitted parameter.

3. **Polyatomic extension:** Generalize the vibrational chain to normal modes of polyatomic molecules, each carrying its own SU(2) Morse representation, coupled through Fermi resonances and Darling-Dennison coupling.

4. **Nuclear graph Laplacian eigenvalues:** Investigate whether the eigenvalues of L_{nuc} (not just the diagonal Hamiltonian) encode additional physical information, such as transition selection rules or Franck-Condon-like overlap integrals, analogous to how the electron graph Laplacian eigenvalues encode the Rydberg spectrum.

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